RV College of Engineering

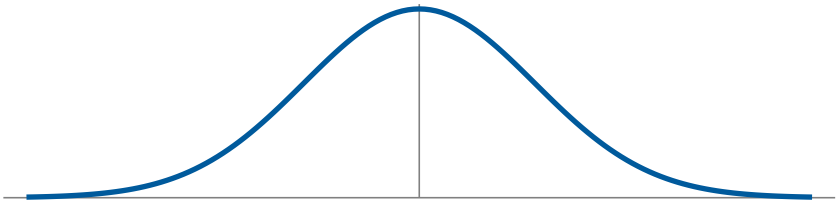
Department of Industrial Engineering and Management

Statistics for Data Analytics

Course Code: IM241AT

Semester IV • Professional Core Course

Unit 3



A Student Compilation for Study and Reference

Semester: IV						
STATISTICS FOR DATA ANALYTICS **Category: Professional Core Course** **(Theory)**						
Course Code	**:**	**IM241AT**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Total Hours	**:**	**45L**		**SEE Duration**	**:**	**3.00 Hours**

Unit-I	**09 Hrs**
Data Summary and Presentation: Data types, tabular and graphical displays: Stem and Leaf diagrams, Histograms, Box plots, Radar diagrams. **Concepts of Probability:** Sample spaces and Events, Interpretations of probability, Addition rules, Conditional probability, Multiplication and Total probability rules, Independence, Bayes Theorem. Numerical Problems	
Unit-II	**09 Hrs**
Concepts of Random Variables: Random Variables, Discrete and continuous random variables. Probability distributions and mass functions, Numerical Problems **Discrete Probability Distributions:** Discrete uniform, Binominal, Poisson, Geometric, Negative binomial, Applications, Numerical Problems.	
Unit-III	**09 Hrs**
Continuous Probability Distributions: Continuous uniform, Normal, Normal approximations, Exponential, Applications, Numerical Problems. **Sampling Distribution:** Sampling distribution, Central Limit Theorem, Sampling distribution of means and derived quantities, Numerical Problems.	
Unit-IV	**09 Hrs**
Simple Linear Regression and Correlation: Empirical models, Simple Linear Regression, Properties of Least square Estimators and Estimation of variances, Common abuses of regression, Prediction of new observations, Correlation, Numerical Problems. Interpretation of graphical output from software packages such as Minitab	
Unit-V	**09 Hrs**
Statistical Inference for a single sample: Hypothesis testing, Inference on the mean of a normal population (variance known and unknown), Inference on the variance of a Normal population, Testing for Goodness of Fit, Tests of association, Numerical Problems **Interval Estimation:** Confidence Intervals on mean (variance known and unknown), and variance of a normal population	

Course Outcomes: After completing the course, the students will be able to	
CO1	Describe and report data set using data analysis, presentation and interpretation techniques to understand various phenomena in the fields of science and engineering.
CO2	Apply various statistical processing techniques to handle a set of data to estimate probabilities.
CO3	Apply an appropriate statistical tool and analyze a specific set of data to estimate and draw conclusions about population parameters
CO4	Draw inferences about population parameters and relations between variables based on analysis of sample data

References Books	
1.	Engineering Statistics, Douglas C. Montgomery, George C. Runger, Norma FarisHubele, 5th Edition, 2011, John Wiley & Sons, Inc., ISBN-13: 978- 0-470-63147-8
2.	Applied statistics and Probability for Engineers, Douglas C Montgomery, George C Runger, Wiley, 4th Edition, 2007, Asia Student Edition, ISBN: 978-81-265-2315-3.
3.	Statistics for Management, Richard I Levin, David S Rubin, 7th Edition, 1997, Prentice Hall India, ISBN: 9780134762920.
4.	Probability and Statistics for Engineers and Scientists, Walpole, Myers, Myers, Ye, 8th Edition, 2007, Pearson Education Inc., ISBN: 978-81-317-1552-9.
5.	Softwares : Microsoft Excel / Minitab / Matlab / R

6.	Online resources: a) http://172.16.44.44/nnptel.html - choose NOC:Introduction to Data Analytics(Course sponsored by Aricent) b) https://onlinecourses.nptel.ac.in/noc22_mg31/preview -Introduction to probability and Statistics c) https://newonlinecourses.science.psu.edu/statprogram/undergraduate-studies d) https://www.khanacademy.org/math/statistics-probability

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
#	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks. **THE SUM OF TWO QUIZZES WILL BE THE FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO tests will be conducted. Each test will be evaluated for 50 Marks, adding upto 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (10), Program specific requirements (10), Video based seminar/presentation/demonstration (20) **ADDING UPTO 40**	**40**
MAXIMUM MARKS FOR THE CIE THEORY		**100**

RUBRIC FOR SEMESTER END EXAMINATION (THEORY)		
Q. NO.	**CONTENTS**	**MARKS**
	PART A	
1	Objective type questions covering entire syllabus	20
	PART B (Maximum of TWO Sub-divisions only)	
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: (Internal Choice)	16
5 & 6	Unit 3: (Internal Choice)	16
7 & 8	Unit 4: (Internal Choice)	16
9 & 10	Unit 5: (Internal Choice)	16
	TOTAL	**100**

A Note on This Compilation

This book is **not** the official RV College of Engineering notes. It is a compilation of student notes together with the handwritten notes provided by **Dr. Ramaa A Nadig**, Department of Industrial Engineering and Management.

All information contained in this book has been drawn from her notes and lectures. The material has been organised, typeset, and arranged to follow the prescribed course syllabus, but it should be treated as a supplementary study aid rather than an official course document.

Every effort has been made to reproduce the source notes faithfully and to preserve their content without addition or deletion. Where the original handwritten working was unclear, the final answers recorded in the notes were treated as authoritative. Readers should always cross-check against the prescribed textbooks and their instructor's guidance.

Compiled from the notes and lectures of Dr. Ramaa A Nadig, Department of Industrial Engineering and Management.

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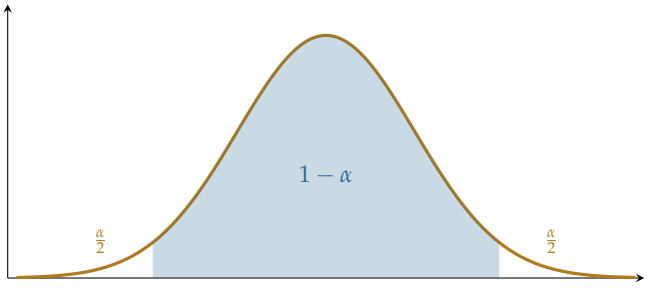
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Unit 3



Preface

This book is a faithful, textbook-format reconstruction of a set of handwritten course notes on the *Estimation of Parameters*, a core topic in engineering statistics. The material has been reorganised into three chapters that follow the natural development of the subject: the general theory of **point estimation**, the two principal **methods of constructing estimators** (moments and maximum likelihood), and the construction of **statistical (confidence) intervals** for a single sample.

Every definition, theorem, worked example, figure, and final numerical answer found in the original notes has been preserved. Connective prose has been added so that each chapter reads as a continuous narrative rather than a list of disconnected results. Definitions appear in blue boxes and the central formulas in amber boxes; all hand-drawn diagrams have been redrawn as clean vector figures. Where the original working was abbreviated, intermediate steps have been supplied so that each derivation reaches the answer recorded in the notes, which remain authoritative.

Chapter 1

Point Estimation

1.1 Statistical Inference and the Estimation Problem

Statistical inference may be divided into two major areas: *parameter estimation* and *hypothesis testing*. As an example of a parameter estimation problem, suppose that a structural engineer is analysing the tensile strength of a component used in an automobile chassis. Because variability in tensile strength is naturally present between individual components—owing to differences in raw-material batches, manufacturing processes, and measurement procedures—the engineer is interested in estimating the mean tensile strength of the component. In practice the engineer will use sample data to compute a number that is, in some sense, a reasonable value (or guess) of the true mean. This number is called a *point estimate*.

Now consider a situation in which two different reaction temperatures t_1 and t_2 can be used in a chemical process. The engineer conjectures that t_1 results in higher yields than t_2. Statistical hypothesis testing is a framework for solving problems of this kind; the present unit concentrates on the first area, estimation.

In inference problems it is convenient to have a general symbol to represent the parameter of interest. The Greek symbol θ (theta) is used to represent the parameter. The objective of point estimation is to select a single number, based on sample data, that is the most plausible value for θ. A numerical value of a sample statistic will be used as the point estimate.

Point Estimator

In general, if X is a random variable with probability distribution $f(x)$, characterised by the unknown parameter θ, and if $X_1, X_2, \ldots, X_n$ is a random sample of size n from X, then the statistic $\hat{\Theta} = h(X_1, X_2, \ldots, X_n)$ is called a **point estimator** of θ. Note that $\hat{\Theta}$ is a random variable, because it is a function of random variables.

Estimation problems occur frequently in engineering. We often need to estimate:

- the mean μ of a single population;
- the variance σ^2 (or standard deviation σ) of a single population;
- the proportion p of items in a population that belong to a class of interest;
- the difference in means of two populations, $\mu_1 - \mu_2$;
- the difference in two population proportions, $p_1 - p_2$.

Reasonable point estimates of these parameters are as follows:

- For μ, the estimate is $\hat{\mu} = \bar{x}$, the sample mean.
- For σ^2, the estimate is $\hat{\sigma}^2 = s^2$, the sample variance.
- For p, the estimate is $\hat{p} = x/n$, the sample proportion, where x is the number of items in a random sample of size n that belong to the class of interest.
- For $\mu_1 - \mu_2$, the estimate is $\hat{\mu}_1 - \hat{\mu}_2 = \bar{x}_1 - \bar{x}_2$, the difference between the sample means of two independent random samples.
- For $p_1 - p_2$, the estimate is $\hat{p}_1 - \hat{p}_2$, the difference between two sample proportions computed from two independent random samples.

1.2 Sampling Distributions and the Central Limit Theorem

Statistical inference is concerned with making decisions about a population based on the information contained in a random sample drawn from that population.

Random Sample

The random variables $X_1, X_2, \ldots, X_n$ are a **random sample** of size n if (a) the X_i are independent random variables and (b) every X_i has the same probability distribution. This property is known as *independent and identically distributed* (i.i.d.).

The observed data is also referred to as a random sample, but the use of the same phrase should not cause any confusion.

Statistic and Sampling Distribution

A **statistic** is any function of the observations in a random sample. The probability distribution of a statistic is called its **sampling distribution**.

Theorem 1.1 (Central Limit Theorem). *If $X_1, X_2, \ldots, X_n$ is a random sample of size n taken from a population (either finite or infinite) with mean μ and finite variance σ^2, and if $\overline{X}$ is the sample mean, then the limiting form of the distribution of*

$$Z = \frac{\overline{X} - \mu}{\sigma / \sqrt{n}}$$

as $n \to \infty$ is the standard normal distribution.

The normal approximation for $\overline{X}$ depends on the sample size n. Figure 1.1 illustrates this dependence using the average score obtained when tossing dice. Part (a) shows the distribution obtained for throws of a single, six-sided true die: the probabilities are equal ($\frac{1}{6}$) for all values. Parts (b)–(e) show the distribution of the average score obtained when tossing 2, 3, 5, and 10 dice respectively. In the case of one die the population is relatively far from normal; the distribution of the average is approximated reasonably well by the normal distribution for sample sizes as small as $n = 5$.

Although the central limit theorem will work well for small samples ($n = 4, 5$) in most cases—particularly where the population is continuous, unimodal, and symmetric—larger samples are required in other situations, depending on the shape of the population. In many cases, if $n \geq 30$, the normal approximation will be satisfactory regardless of the shape of the population. If $n < 30$, the central limit theorem will work if the distribution of the population is not severely non-normal.

1.3 General Concepts of Point Estimation

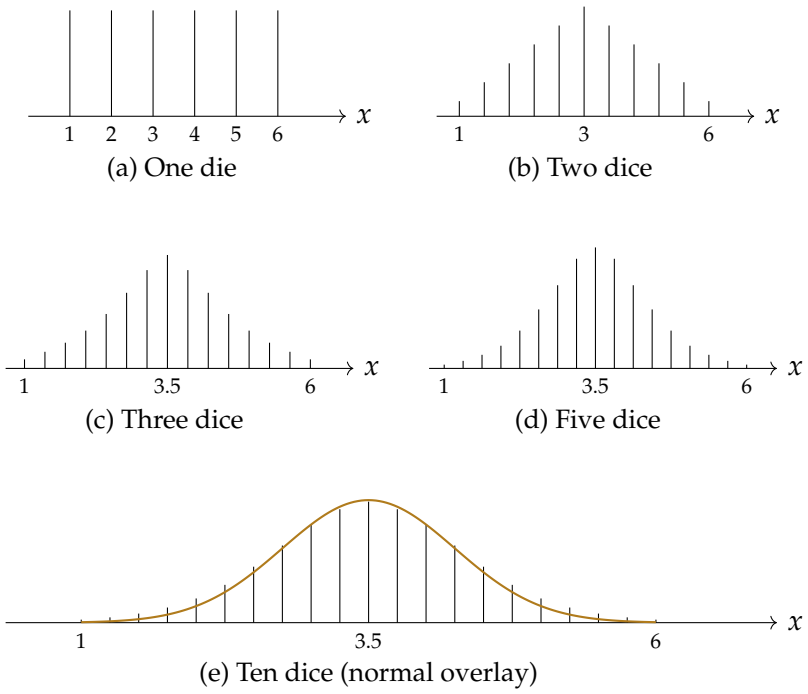
1.3.1 Unbiased Estimators

Unbiased Estimator

The point estimator $\hat{\Theta}$ is an **unbiased estimator** for the parameter θ if

$$\mathbb{E}(\hat{\Theta}) = \theta.$$

If the estimator is not unbiased, then the difference $\mathbb{E}(\hat{\Theta}) - \theta$ is called the **bias** of the



(a) One die (b) Two dice

(c) Three dice (d) Five dice

(e) Ten dice (normal overlay)

Figure 1.1: The distribution of the average score obtained when tossing $n = 1, 2, 3, 5$ and 10 dice. As n grows, the sampling distribution of the average approaches the normal shape.

estimator $\hat{\Theta}$. When an estimator is unbiased, the bias is zero: $\mathbb{E}(\hat{\Theta}) - \theta = 0$.

Example 1.1 (Sample mean and sample variance are unbiased)**.** Suppose that X is a random variable with mean μ and variance σ^2. Let $X_1, X_2, \ldots, X_n$ be a random sample of size n from the population represented by X. Show that the sample mean $\overline{X}$ and the sample variance S^2 are unbiased estimators of μ and σ^2 respectively.

Sample mean. We know that $\mathbb{E}(\overline{X}) = \mu$; therefore the sample mean $\overline{X}$ is the unbiased estimator of the population mean μ.

Sample variance. Consider

$$\mathbb{E}(S^2) = \mathbb{E}\left[\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}\right] = \frac{1}{n-1}\,\mathbb{E}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Expanding the square,

$$\mathbb{E}(S^2) = \frac{1}{n-1}\,\mathbb{E}\sum_{i=1}^{n}\left(X_i^2 + \overline{X}^2 - 2\overline{X}X_i\right) = \frac{1}{n-1}\,\mathbb{E}\left[\sum X_i^2 - 2\overline{X}\sum X_i + n\overline{X}^2\right].$$

Since $\sum X_i = n\overline{X}$, the middle term becomes $2\overline{X}\cdot n\overline{X} = 2n\overline{X}^2$, so

$$\begin{aligned}\mathbb{E}(S^2) &= \frac{1}{n-1}\,\mathbb{E}\left[\sum X_i^2 - 2n\overline{X}^2 + n\overline{X}^2\right] = \frac{1}{n-1}\,\mathbb{E}\left[\sum_{i=1}^{n} X_i^2 - n\overline{X}^2\right] \\ &= \frac{1}{n-1}\left[\sum_{i=1}^{n}\mathbb{E}(X_i^2) - n\,\mathbb{E}(\overline{X}^2)\right]. \qquad (1)\end{aligned}$$

Using the identities $\mathbb{E}(\sum X_i) = \sum \mathbb{E}(X_i)$, $\mathbb{E}(cX) = c\,\mathbb{E}(X)$ and $\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$, we have

$$\mathbb{E}(X_i^2) = \mu^2 + \sigma^2, \qquad \mathbb{E}(\overline{X}^2) = \mathrm{Var}(\overline{X}) + [\mathbb{E}(\overline{X})]^2 = \mu^2 + \frac{\sigma^2}{n}.$$

Substituting into (1),

$$\begin{aligned}\mathbb{E}(S^2) &= \frac{1}{n-1}\left[\sum_{i=1}^{n}(\mu^2+\sigma^2) - n\left(\mu^2 + \frac{\sigma^2}{n}\right)\right] \\ &= \frac{1}{n-1}\left[n\mu^2 + n\sigma^2 - n\mu^2 - \sigma^2\right] = \frac{\sigma^2(n-1)}{n-1} = \sigma^2.\end{aligned}$$

Therefore the sample variance S^2 is an unbiased estimator of the population variance σ^2. Although S^2 is unbiased for σ^2, S is a *biased* estimator of σ; for large samples the bias is very small.

1.3.2 Variance of a Point Estimator

Suppose there are two unbiased estimators of θ. This indicates that the distribution of each is centred at the true value of θ; however, the variances of these distributions may be different (Figure 1.2). A logical principle of estimation, when selecting among several estimators, is to choose the estimator that has minimum variance.

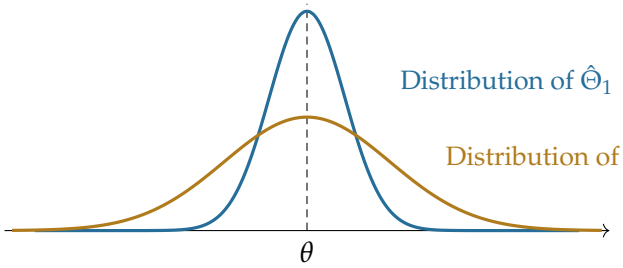


Figure 1.2: Two unbiased estimators of θ. Both distributions are centred at the true value θ, but $\hat{\Theta}_1$ has the smaller variance and is therefore preferred.

Minimum Variance Unbiased Estimator (MVUE)

If we consider all unbiased estimators of θ, the one with the smallest variance is called the **minimum variance unbiased estimator** (MVUE). If $X_1, X_2, \ldots, X_n$ is a random sample of size n from a normal distribution with mean μ and variance σ^2, then the sample mean $\overline{X}$ is the MVUE for μ.

1.3.3 Standard Error of an Estimator

Standard Error

The **standard error** of an estimator $\hat{\Theta}$ is its standard deviation, given by $\sigma_{\hat{\Theta}} = \sqrt{V(\hat{\Theta})}$. If the standard error involves unknown parameters that can themselves be estimated, substitution of those values into $\sigma_{\hat{\Theta}}$ produces an **estimated standard error**, denoted $\hat{\sigma}_{\hat{\Theta}}$.

Suppose we are sampling from a normal distribution with mean μ and variance σ^2. The distribution of $\overline{X}$ is normal with mean μ and variance σ^2/n, so the standard error of $\overline{X}$ is

Standard Error of the Sample Mean

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}.$$

In cases in which the point estimator is not normally distributed, we can state that so long as the estimator is unbiased, the estimate of the parameter will deviate from the true value by as much as four standard errors at most 6% of the time. This is called *Chebyshev's inequality*.

1.3.4 Bias: Two Worked Examples

Example 1.2 (Bias of $\hat{\mu} = \overline{X}$ for a Poisson mean)**.** Determine the bias of the estimator $\hat{\mu} = \overline{X}$ of the parameter μ from a Poisson(μ) distribution.

$$\begin{aligned}\text{Bias}(\hat{\mu}) &= \mathbb{E}(\hat{\mu}) - \mu = \mathbb{E}(\overline{X}) - \mu = \mathbb{E}\left(\tfrac{1}{n}\sum X_i\right) - \mu = \tfrac{1}{n}\sum \mathbb{E}(X_i) - \mu \\ &= \tfrac{1}{n}\sum \mu - \mu = \tfrac{1}{n}\, n\mu - \mu = \mu - \mu = 0.\end{aligned}$$

The estimator is unbiased.

Example 1.3 (Bias of $\hat{p} = X/n$ for a binomial proportion)**.** Determine the bias of the estimator $\hat{p} = X/n$ of the parameter p from a Bin(n, p) distribution. Here $X \sim \text{Bin}(n, p)$ so $\mathbb{E}(X) = np$.

$$\text{Bias}(\hat{p}) = \mathbb{E}(\hat{p}) - p = \mathbb{E}\left(\tfrac{X}{n}\right) - p = \tfrac{1}{n}\mathbb{E}(X) - p = \tfrac{1}{n}\, np - p = p - p = 0.$$

The estimator is unbiased.

1.3.5 Mean Squared Error of an Estimator

Mean Squared Error

The **mean squared error** of an estimator $\hat{\Theta}$ of the parameter θ is

$$\text{MSE}(\hat{\Theta}) = \mathbb{E}(\hat{\Theta} - \theta)^2.$$

It can be rewritten as

$$\text{MSE}(\hat{\Theta}) = \mathbb{E}\left[\hat{\Theta} - \mathbb{E}(\hat{\Theta})\right]^2 + \left[\theta - \mathbb{E}(\hat{\Theta})\right]^2 = V(\hat{\Theta}) + (\text{Bias})^2,$$

that is, the MSE is equal to the variance of the estimator plus the squared bias.

MSE Decomposition

$$\text{MSE}(\hat{\Theta}) = V(\hat{\Theta}) + \left[\text{Bias}(\hat{\Theta})\right]^2.$$

The mean squared error is an important criterion for comparing two estimators. Consider two estimators $\hat{\Theta}_1$ and $\hat{\Theta}_2$ of the parameter θ, and let $\text{MSE}(\hat{\Theta}_1)$ and $\text{MSE}(\hat{\Theta}_2)$ be their mean squared errors. The **relative efficiency** of $\hat{\Theta}_2$ to $\hat{\Theta}_1$ is defined as

$$\frac{\text{MSE}(\hat{\Theta}_1)}{\text{MSE}(\hat{\Theta}_2)}.$$

If this relative efficiency is less than 1, we would conclude that $\hat{\Theta}_1$ is a more efficient estimator of θ than $\hat{\Theta}_2$.

Sometimes we find that biased estimators are preferable to unbiased estimators because they have smaller mean squared error; that is, we may be able to reduce the variance of the estimator considerably by introducing a relatively small amount of bias. An estimator that has a mean squared error less than or equal to the mean squared error of any other estimator, for all values of the parameter θ, is called an **optimal estimator** of θ.

1.4 Worked Problems

Example 1.4 (Comparing two sample means of different sizes)**.** Suppose we have a random sample of size $2n$ from a population denoted by X, with $\mathbb{E}(X) = \mu$ and $V(X) = \sigma^2$. Let

$$\overline{X}_1 = \frac{1}{2n}\sum_{i=1}^{2n} X_i \qquad \text{and} \qquad \overline{X}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i$$

be two estimators of μ. Which is the better estimator of μ? Explain.

Expected values.

$$\mathbb{E}(\overline{X}_1) = \frac{1}{2n}\sum_{i=1}^{2n}\mathbb{E}(X_i) = \frac{1}{2n}(2n)\mu = \mu, \qquad \mathbb{E}(\overline{X}_2) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i) = \frac{1}{n}(n)\mu = \mu.$$

Both estimators have the same expected value μ, so both are unbiased; the better estimator is the one with the smaller variance.

Variances.

$$V(\overline{X}_1) = V\left(\frac{1}{2n}\sum_{i=1}^{2n} X_i\right) = \frac{1}{(2n)^2}\sum_{i=1}^{2n} V(X_i) = \frac{1}{4n^2}(2n)\sigma^2 = \frac{\sigma^2}{2n},$$

$$V(\overline{X}_2) = V\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} V(X_i) = \frac{1}{n^2}\,n\sigma^2 = \frac{\sigma^2}{n}.$$

Since $V(\overline{X}_1) = \sigma^2/2n$ is smaller than $V(\overline{X}_2) = \sigma^2/n$, the estimator $\overline{X}_1$ is the better estimator.

Example 1.5 (Unbiasedness, best estimator, and relative efficiency)**.** Let $X_1, X_2, \ldots, X_7$ denote a random sample from a population having mean μ and variance σ^2. Consider the following two estimators of μ:

$$\hat{\Theta}_1 = \frac{X_1 + X_2 + \cdots + X_7}{7}, \qquad \hat{\Theta}_2 = \frac{2X_1 - X_6 + X_4}{2}.$$

(a) Is either estimator unbiased? **(b)** Which estimator is best, and in what sense? Calculate the relative efficiency of the two estimators.

(a) Unbiasedness.

$$\mathbb{E}(\hat{\Theta}_1) = \mathbb{E}\left(\frac{X_1 + \cdots + X_7}{7}\right) = \frac{1}{7}\mathbb{E}(X_1) + \cdots + \frac{1}{7}\mathbb{E}(X_7) = \frac{1}{7}\mu + \cdots + \frac{1}{7}\mu = \mu,$$

$$\mathbb{E}(\hat{\Theta}_2) = \mathbb{E}\left(\frac{2X_1 - X_6 + X_4}{2}\right) = \mathbb{E}(X_1) - \tfrac{1}{2}\mathbb{E}(X_6) + \tfrac{1}{2}\mathbb{E}(X_4) = \mu - \tfrac{1}{2}\mu + \tfrac{1}{2}\mu = \mu.$$

Hence $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are both unbiased estimators of μ.

(b) Variances.

$$\mathrm{Var}(\hat{\Theta}_1) = \mathrm{Var}\left(\frac{X_1 + \cdots + X_7}{7}\right) = \frac{1}{49}\left[\mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_7)\right] = \frac{1}{49}\,(7\sigma^2) = \frac{1}{7}\sigma^2.$$

$$\text{Var}(\hat{\Theta}_2) = \text{Var}\left(\frac{2X_1 - X_6 + X_4}{2}\right) = \frac{1}{4}\text{Var}(2X_1) + \frac{1}{4}\text{Var}(X_6) + \frac{1}{4}\text{Var}(X_4).$$

Since $\text{Var}(2X_1) = 4\,\text{Var}(X_1) = 4\sigma^2$,

$$\text{Var}(\hat{\Theta}_2) = \frac{4}{4}\sigma^2 + \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2 = \sigma^2 + \tfrac{1}{4}\sigma^2 + \tfrac{1}{4}\sigma^2 = \frac{3}{2}\sigma^2.$$

The best estimator is the one with the smaller variance, so $\hat{\Theta}_1$ is the best estimator. The relative efficiency of the two estimators is

$$\frac{\text{Var}(\hat{\Theta}_2)}{\text{Var}(\hat{\Theta}_1)} = \frac{\frac{3}{2}\sigma^2}{\frac{1}{7}\sigma^2} = \frac{21}{2}.$$

Example 1.6 (MSE comparison of a biased and an unbiased estimator)**.** Suppose that $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are estimators of the parameter θ. We know that $\mathbb{E}(\hat{\Theta}_1) = \theta$, $\mathbb{E}(\hat{\Theta}_2) = \theta/2$, $V(\hat{\Theta}_1) = 10$, and $V(\hat{\Theta}_2) = 4$. Which estimator is best, and in what sense?

Since the means of the two estimators differ, the mean squared errors also differ.

$$\text{MSE}(\hat{\Theta}_1) = V(\hat{\Theta}_1) + \left[\mathbb{E}(\hat{\Theta}_1) - \theta\right]^2 = 10 + (\theta - \theta)^2 = 10,$$
$$\text{MSE}(\hat{\Theta}_2) = V(\hat{\Theta}_2) + \left[\mathbb{E}(\hat{\Theta}_2) - \theta\right]^2 = 4 + \left(\tfrac{\theta}{2} - \theta\right)^2 = 4 + \frac{\theta^2}{4}.$$

The relative efficiency is

$$\frac{\text{MSE}(\hat{\Theta}_1)}{\text{MSE}(\hat{\Theta}_2)} = \frac{10}{4 + \frac{\theta^2}{4}} = \frac{40}{16 + \theta^2}.$$

- If $\theta^2 = 24$, the relative efficiency is $1 \Rightarrow \hat{\Theta}_1$ and $\hat{\Theta}_2$ are equally efficient.
- If $\theta^2 > 24$, the relative efficiency is $< 1 \Rightarrow \hat{\Theta}_1$ is more efficient.
- If $\theta^2 < 24$, the relative efficiency is $> 1 \Rightarrow \hat{\Theta}_2$ is more efficient.

Example 1.7 (Pull-off force for connectors)**.** Data on the pull-off force (pounds) for connectors used in an automobile engine application are as follows:

> 79.3, 75.1, 78.2, 74.1, 73.9, 75.0, 77.6, 77.3, 73.8, 74.6, 75.5, 74.0, 74.7, 75.9, 72.9, 73.8, 74.2, 78.1, 75.4, 76.3, 75.3, 76.2, 74.9, 78.0, 75.1, 76.8.

(a) Calculate a point estimate of the mean pull-off force of all connectors in the population. State which estimator you used and why. **(b)** Calculate a point estimate of the pull-off force value that separates the weakest 50% of the connectors from the strongest 50%. **(c)** Calculate point estimates of the population variance and the population standard deviation. **(d)** Calculate the standard error of the point estimate found in part (a). **(e)** Calculate a point estimate of the proportion of all connectors whose pull-off force is less than 73 pounds.

(a) Mean. The sample mean is

$$\overline{X} = \frac{1966}{26} = 75.62 \text{ pounds}.$$

The population mean μ is estimated by the sample mean $\overline{X} = 75.62$ pounds, because it is known that the sample mean is an unbiased estimator of the population mean.

(b) Median. Sorting the 26 observations,

72.9, 73.8, 73.8, 73.9, 74.0, 74.1, 74.2, 74.6, 74.7, 74.9, 75.0, 75.1, 75.1, 75.3, 75.4, 75.5, 75.9, 76.2, 76.3, 76.8, 77.3, 77.6, 78.0, 78.1, 78.2, 79.3.

The sample median is 75.2 pounds, which is the estimate of the population median.

(c) Variance and standard deviation. Good point estimates of the population variance and standard deviation are the corresponding sample quantities:

$$\widehat{\sigma^2} = s^2 = 2.7382 \text{ pounds}^2, \qquad \hat{\sigma} = s = 1.6547 \text{ pounds}.$$

(d) Standard error. The standard error of the point estimate for the mean pull-off force is

$$\text{SE}(\overline{X}) = \hat{\sigma}_{\overline{X}} = \frac{s}{\sqrt{n}} = \frac{1.6547}{\sqrt{26}} = 0.3245.$$

The standard error is about 0.4% of the sample mean, which means a relatively precise point estimate of pull-off force has been obtained. If pull-off force is assumed normally distributed, then $2\hat{\sigma}_{\overline{X}} = 2(0.3245) = 0.6490$, so we are confident that the true mean pull-off force lies within 75.62 ± 0.649, that is, $(74.97, 76.26)$ pounds.

(e) Proportion below 73 pounds. A point estimate of the proportion of all connectors whose pull-off force is less than 73 pounds is

$$\hat{p} = \frac{x}{n} = \frac{\text{\# specimens less than 73}}{\text{total}} = \frac{1}{26} = 0.038462.$$

Chapter 2

Methods of Point Estimation

The previous chapter described desirable *properties* of an estimator. This chapter describes two general *methods* for actually constructing point estimators: the method of moments and the method of maximum likelihood.

2.1 Method of Moments

Moments

Let $X_1, X_2, \ldots, X_n$ be a random sample from the probability distribution $f(x)$, where $f(x)$ can be a discrete probability mass function or a continuous probability density function. The kth **population moment** (or distribution moment) is $\mathbb{E}(X^k)$, $k = 1, 2, \ldots$. The corresponding kth **sample moment** is $\frac{1}{n}\sum_{i=1}^{n} X_i^k$, $k = 1, 2, \ldots$.

Moment Estimators

Let $X_1, X_2, \ldots, X_n$ be a random sample from either a probability mass function or a probability density function with m unknown parameters $\theta_1, \theta_2, \ldots, \theta_m$. The **moment estimators** $\hat{\Theta}_1, \hat{\Theta}_2, \ldots, \hat{\Theta}_m$ are found by equating the first m population moments to the first m sample moments and solving the resulting equations for the unknown parameters.

In practice the method of moments proceeds in three steps:

1. Calculate the first few sample moments, $m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i$ and $m_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2$.
2. Calculate the corresponding population moments $\mathbb{E}(X^k) = \int_{-\infty}^{\infty} x^k f(x; \theta_1, \theta_2)\, dx$; the results are expressed in terms of θ_1 and θ_2.
3. Solve the simultaneous equations $\mathbb{E}(X) = m_1$, $\mathbb{E}(X^2) = m_2$ for the unknown parameters. The solutions are the method-of-moments estimators.

2.2 Method of Maximum Likelihood

Maximum Likelihood Estimator

Suppose that X is a random variable with probability distribution $f(x; \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \ldots, x_n$ be the observed values in a random sample of size n. Then the **likelihood function** of the sample is

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdots f(x_n; \theta).$$

The likelihood function is now a function of only the unknown parameter θ. The **maximum likelihood estimator** (MLE) of θ is the value of θ that maximises the likelihood function $L(\theta)$.

Properties of a Maximum Likelihood Estimator

Under very general and not restrictive conditions, when the sample size n is large and if $\hat{\Theta}$ is the maximum likelihood estimator of the parameter θ, then:

1. $\hat{\Theta}$ is an approximately unbiased estimator for θ, i.e. $\mathbb{E}(\hat{\Theta}) \approx \theta$;
2. the variance of $\hat{\Theta}$ is nearly as small as the variance that could be obtained with any other estimator; and
3. $\hat{\Theta}$ has an approximate normal distribution.

2.3 Maximum Likelihood: Worked Problems

Example 2.1 (MLE for a Poisson distribution)**.** Consider the Poisson distribution with parameter λ. Find the maximum likelihood estimator of λ, based on a random sample of size n. The probability mass function is

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

The likelihood function for a random sample of size n is

$$L(\lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}\,\lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}.$$

Applying the logarithm to both sides,

$$\log L(\lambda) = -n\lambda + \left(\sum_{i=1}^n x_i\right)\log\lambda - \log\left(\prod_{i=1}^n x_i!\right).$$

Differentiating with respect to λ and equating to zero,

$$\frac{d\log L(\lambda)}{d\lambda} = -n + \frac{\sum_{i=1}^n x_i}{\lambda} = 0 \implies \frac{\sum_{i=1}^n x_i}{\lambda} = n \implies \lambda = \frac{\sum_{i=1}^n x_i}{n}.$$

$$\hat{\lambda} = \overline{X}.$$

Example 2.2 (MLE for a geometric distribution)**.** Let X be a geometric random variable with parameter p. Find the maximum likelihood estimator of p based on a random sample of size n. The density is

$$f(x) = (1-p)^{x-1}p, \qquad x = 1, 2, \ldots$$

The likelihood function is

$$L(p) = \prod_{i=1}^{n} p\,(1-p)^{x_i-1} = p^n \prod_{i=1}^{n}(1-p)^{x_i-1}.$$

Taking logarithms,

$$\log L(p) = n\log p + \left(\sum_{i=1}^n x_i - n\right)\log(1-p).$$

Differentiating with respect to p and equating to zero,

$$\frac{d\log L(p)}{dp} = \frac{n}{p} + \left(\sum_{i=1}^n x_i - n\right)\frac{-1}{1-p} = 0 \implies \frac{n}{p} = \frac{\sum_{i=1}^n x_i - n}{1-p}.$$

Rearranging,

$$\frac{1}{p} = 1 + \frac{\sum_{i=1}^n x_i - n}{n} = \frac{\sum_{i=1}^n x_i}{n},$$

$$\hat{p} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\overline{X}}.$$

Example 2.3 (MLE for the density $(\theta+1)x^\theta$)**.** Let X be a random variable with the probability distribution

$$f(x) = \begin{cases} (\theta+1)x^\theta, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the maximum likelihood estimator of θ, based on a random sample of size n. The likelihood function is

$$L(\theta) = \prod_{i=1}^{n} f(x_i) = (\theta+1)^n \left[x_1 x_2 \cdots x_n\right]^\theta.$$

Taking logarithms and differentiating with respect to θ,

$$\frac{d}{d\theta} \log L(\theta) = \frac{d}{d\theta}\left[n \log(\theta+1) + \theta \textstyle\sum_{i=1}^{n} \log x_i\right] = \frac{n}{\theta+1} + \sum_{i=1}^{n} \log x_i = 0.$$

Hence

$$\sum_{i=1}^{n} \log x_i = \frac{-n}{\theta+1} \implies \theta + 1 = \frac{-n}{\sum_{i=1}^{n} \log x_i} \implies \hat{\Theta} = -\left[1 + \frac{n}{\sum_{i=1}^{n} \log x_i}\right].$$

2.4 Method of Moments: Worked Problems

Example 2.4 (Poisson mean by the method of moments; significance of the MSE)**.** Ten values collected at random from a Poisson distribution with mean μ are

$$10,\ 11,\ 15,\ 17,\ 9,\ 13,\ 18,\ 11,\ 15,\ 16.$$

Estimate the mean parameter μ using the method of moments. What significance does the mean squared error hold for an estimator, and how is it used to evaluate an estimator?

Here $X \sim \text{Poi}(\mu)$, and we know $\mathbb{E}(X) = \mu$. The method of moments equates $\mathbb{E}(X)$ to the first sample moment:

$$\mathbb{E}(X) = \overline{X} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{10}\sum_{i=1}^{10} x_i.$$

Therefore

$$\hat{\mu} = \frac{1}{10}\sum x_i = \frac{135}{10} = 13.5.$$

The mean squared error combines the effect of the bias and the variance of an estimator; the smaller the MSE, the better the estimator:

$$\text{MSE} = (\text{Bias})^2 + \text{Var}\,.$$

It is possible that, of two estimators, one with higher bias or higher variance may have a lower MSE.

Example 2.5 (Method-of-moments estimator for a binomial proportion)**.** The random variable X represents the number of defective parts in a lot of 90 parts. A sample of 25 lots gives x_i,

$i = 1, 2, \ldots, 25$, where each x_i is the number of defective parts in the ith lot. Derive the method-of-moments estimator for the proportion p of a part being defective. Here $X \sim \text{Binomial}(90, p)$, where p is the proportion defective, so $\mathbb{E}(X) = 90p$. Equating $\mathbb{E}(X)$ to the sample moment,

$$90p = \frac{1}{25}\sum_{i=1}^{25} x_i \implies \hat{p} = \frac{1}{90}\left[\frac{1}{25}\sum_{i=1}^{25} x_i\right] = \frac{\overline{X}}{90}.$$

The population mean (np) of a binomial may be estimated in several ways: $\overline{X}$, $\frac{1}{2}\big(\max(x_i) + \min(x_i)\big)$, or $\bar{y}$, where $\bar{y}$ is computed from the 23 values y_i obtained after excluding $\min(x_i)$ and $\max(x_i)$. The corresponding mean squared errors are

$$\begin{aligned}
\text{MSE}(\overline{X}) &= [\text{Bias}(\overline{X})]^2 + \text{Var}(\overline{X}) = 0 + \frac{90p(1-p)}{25}, \\
\text{MSE}\Big(\tfrac{\max(x_i)+\min(x_i)}{2}\Big) &= 0 + \frac{90p(1-p)}{2}, \\
\text{MSE}(\bar{y}) &= 0 + \frac{90p(1-p)}{23}.
\end{aligned}$$

Therefore $\overline{X}$ is best, since its MSE is minimum. In summary: bias is used for unbiasedness, standard error is used to determine low spread, and the MSE is used for checking consistency.

Example 2.6 (Method-of-moments estimators for the normal distribution)**.** Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution with parameters μ and σ^2. For the normal distribution,

$$\mathbb{E}(X) = \mu, \qquad \mathbb{E}(X^2) = \mu^2 + \sigma^2,$$

since $\mathbb{E}(X - \mu) = 0$ and $\mathbb{E}[(X-\mu)^2] = \sigma^2$. Equating $\mathbb{E}(X) = \overline{X}$ and $\mathbb{E}(X^2) = \frac{1}{n}\sum_{i=1}^n X_i^2$ gives

$$\mu = \overline{X}, \qquad \mu^2 + \sigma^2 = \frac{1}{n}\sum_{i=1}^n X_i^2.$$

Solving these equations,

$$\hat{\mu} = \overline{X}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n X_i^2 - \overline{X}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X})^2.$$

Notice that the moment estimator of σ^2 is *not* an unbiased estimator (it divides by n rather than $n-1$).

Example 2.7 (Discrete distribution depending on θ)**.** Suppose that X is a discrete random variable with the following distribution, where $0 \le \theta \le 1$ is a parameter:

x	0	1	2	3
$P(x)$	$\frac{2\theta}{3}$	$\frac{\theta}{3}$	$\frac{2(1-\theta)}{3}$	$\frac{1-\theta}{3}$

The following 10 independent observations were taken from such a distribution: $3, 0, 2, 1, 3, 2, 1, 0, 2, 1$. Use the method of moments to find the estimate of θ.

The theoretical mean value is

$$\mathbb{E}(X) = \sum_{x=0}^{3} xP(x) = 0 \cdot \tfrac{2\theta}{3} + 1 \cdot \tfrac{\theta}{3} + 2 \cdot \tfrac{2(1-\theta)}{3} + 3 \cdot \tfrac{1-\theta}{3} = \frac{7}{3} - 2\theta.$$

The sample mean value is

$$\overline{X} = \frac{3+0+2+1+3+2+1+0+2+1}{10} = 1.5.$$

We need to solve $\frac{7}{3} - 2\theta = 1.5$, and finally obtain

$$2\theta = \frac{7}{3} - 1.5 \implies \theta = \frac{7 - 4.5}{3 \times 2} = \frac{2.5}{6} = \frac{5}{12}.$$

$$\hat{\Theta} = \frac{5}{12}.$$

Example 2.8 (Method of moments for a uniform density)**.** Use the method of moments to estimate the parameter θ for the uniform density $f(x \mid \theta) = \frac{1}{\theta}$ with $0 \leq x \leq \theta$, based on a random sample $X_1, \ldots, X_n$. The theoretical moment is

$$\mathbb{E}(X) = \int_0^\theta x \cdot \frac{1}{\theta}\, dx = \frac{x^2}{2\theta}\Big|_0^\theta = \frac{\theta^2}{2\theta} - 0 = \frac{\theta}{2}.$$

The first sample moment is $\overline{X}$. Equating both,

$$\mathbb{E}(X) = \overline{X} \implies \frac{\theta}{2} = \overline{X} \implies \hat{\Theta} = 2\overline{X} = \frac{2}{n}\sum_{i=1}^{n} x_i,$$

which is the method-of-moments estimate.

A Catalogue of Moment Estimators

The same procedure yields the moment estimators of several standard distributions.

Example 2.9 (Bernoulli distribution)**.** For $X \sim \text{Bern}(p)$, $P_X(x) = p^x(1-p)^{1-x}$. The population mean is $\mathbb{E}(X) = p$ and the sample mean is $\overline{X} = \frac{1}{N}\sum_{i=1}^{N} x_i$, so $\hat{p} = \overline{X} = \frac{1}{N}\sum_{i=1}^{N} x_i$.

Example 2.10 (Geometric distribution)**.** For $X \sim \text{Geo}(p)$, $P_X(x) = p(1-p)^{x-1}$, with $\mathbb{E}(X) = \frac{1}{p}$. Equating to $\overline{X} = \frac{1}{N}\sum_{i=1}^{N} x_i$ gives $\hat{p} = \dfrac{1}{\overline{X}} = \dfrac{N}{\sum_{i=1}^{N} x_i}$.

Example 2.11 (Poisson distribution)**.** For $X \sim \text{Poisson}(\lambda)$, $P_X(x) = \dfrac{e^{-\lambda}\lambda^x}{x!}$, $x \geq 0$, with $\mathbb{E}(X) = \lambda$. Therefore $\hat{\lambda} = \overline{X} = \frac{1}{N}\sum_{i=1}^{N} x_i$.

Example 2.12 (Chi-square distribution)**.** For the chi-square distribution χ^2_p,

$$f_X(x) = \frac{1}{\Gamma\left(\frac{p}{2}\right) 2^{p/2}}\, x^{p/2-1} e^{-x/2},$$

with $\mathbb{E}(X) = p$. The moment estimator of the degrees-of-freedom parameter is $\hat{p} = \overline{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$.

Example 2.13 (Likelihood function of the exponential distribution)**.** Consider the exponential distribution with parameter λ. Derive the likelihood function of a random sample of size n. The likelihood function of the exponential distribution is

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}.$$

Taking the natural logarithm,

$$\ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} x_i.$$

Differentiating with respect to λ and equating to zero,

$$\frac{d \ln L(\lambda)}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0 \implies \frac{n}{\lambda} = \sum_{i=1}^{n} x_i \implies \hat{\lambda} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\overline{X}}.$$

$$\hat{\lambda} = \frac{1}{\overline{X}}.$$

Thus the maximum likelihood estimator of λ is the reciprocal of the sample mean.

Example 2.14 (A density with constant c: four parts)**.** Consider the probability density function

$$f(x) = c(1 + \theta x), \qquad -1 \le x \le 1.$$

(a) Find the value of the constant c. **(b)** What is the moment estimator for θ? **(c)** Show that $\hat{\Theta} = 3\overline{X}$ is an unbiased estimator for θ. **(d)** Find the maximum likelihood estimator for θ.

(a) A valid probability density function satisfies $\int f(x)\,dx = 1$:

$$\int_{-1}^{1} c(1 + \theta x)\,dx = c\left(x + \frac{\theta x^2}{2}\right)\Big|_{-1}^{1} = c\left(1 + \frac{\theta}{2}\right) - c\left(-1 + \frac{\theta}{2}\right) = 1.$$

Hence $c + \dfrac{c\theta}{2} + c - \dfrac{c\theta}{2} = 1 \Rightarrow 2c = 1 \Rightarrow c = \dfrac{1}{2}$.

(b) The moment estimator for θ:

$$\hat{\mu} = \int_{-1}^{1} x\, f(x)\,dx = \int_{-1}^{1} x \cdot \tfrac{1}{2}(1 + \theta x)\,dx = \frac{1}{2}\int_{-1}^{1} (x + \theta x^2)\,dx = \frac{1}{2}\left[\frac{x^2}{2} + \frac{\theta x^3}{3}\right]_{-1}^{1}.$$

The first term vanishes (it is odd), leaving

$$\hat{\mu} = \frac{1}{2}\left[0 + \frac{\theta}{3}(1^3 + 1^3)\right] = \frac{1}{2} \cdot \frac{2\theta}{3} = \frac{\theta}{3}.$$

$$\hat{\mu} = \frac{\theta}{3}.$$

(c) The sample estimator of the mean is $\overline{X}$, and $\hat{\mu} = \dfrac{\theta}{3}$, so $\theta = 3\hat{\mu} \Rightarrow \hat{\Theta} = 3\overline{X}$. To prove $\hat{\Theta} = 3\overline{X}$ is unbiased we must show $\mathbb{E}(3\overline{X}) = \theta$. Since the sample mean $\overline{X}$ is an unbiased estimator of the population mean $\hat{\mu}$,

$$\mathbb{E}(3\overline{X}) = 3\,\mathbb{E}(\overline{X}) = 3\hat{\mu} = 3 \cdot \frac{\theta}{3} = \theta.$$

Hence the moment estimator $3\overline{X}$ is an unbiased estimator of θ.

(d) Maximum likelihood estimator. With $f(x) = \frac{1}{2}(1+\theta x)$,

$$L(\theta) = \prod_{i=1}^{n} f(x_i) = \frac{(1+\theta x_1)(1+\theta x_2)\cdots(1+\theta x_n)}{2^n} = \frac{1}{2^n}\prod_{i=1}^{n}(1+\theta x_i).$$

Taking logarithms,

$$\log L(\theta) = -n\log 2 + \sum_{i=1}^{n}\log(1+\theta x_i).$$

Differentiating,

$$\frac{d}{d\theta}\log L(\theta) = 0 + \sum_{i=1}^{n}\frac{x_i}{1+\theta x_i}.$$

Equating this to zero gives

$$\sum_{i=1}^{n}\frac{x_i}{1+\theta x_i} = 0.$$

By observation the value of θ that maximises the likelihood is $\max(x_i)$.

2.5 Commonly Used Estimators and Criteria

Commonly used estimators are as follows:

1. Maximum likelihood estimators
2. Bayes estimators
3. Method of moments estimators
4. Least squares
5. Minimum mean squared error (MMSE) or Bayes least squared error (BLSE)
6. Minimum variance unbiased estimator (MVUE)
7. Best linear unbiased estimator (BLUE)
8. Nonlinear system identification
9. Unbiased estimators
10. Markov chain Monte Carlo
11. Particle filter
12. Kalman filter
13. Wiener filter

Several criteria are proposed for estimators:

1. **Unbiased estimator** — the long-run average of the estimate equals the true parameter value.
2. **Consistent estimator** — with more and more observations the estimate gets nearer to the true parameter value.
3. **Efficient estimator** — has small variability.
4. **Minimum variance** among all unbiased estimators.

An estimator Y for a parameter θ is unbiased if $\mathbb{E}(Y) = \theta$. Intuitively, Y is unbiased if its long-term average value is equal to θ. In the normal population model, $\overline{X}$ is an unbiased estimator of μ: this is because $\mathbb{E}(X_i) = \mu$ for each i, and hence $\mathbb{E}(\overline{X}) = \mu$. Also S^2 is an unbiased estimator of σ^2. On the other hand, if you put Y as the largest of the observations, then Y is not an unbiased estimator. It appears that, after all, this maximum is one of the observations, and each observation has expected value μ, so Y might be expected to have the same property—but it does not. Indeed, if the sample size is at least two, then $\mathbb{E}(Y) > \mu$.

Consistency

An estimator is **consistent** if it gets closer and closer to the parameter value as the sample size increases, i.e. an estimator Y for a parameter θ is consistent if for any $\epsilon > 0$ we have $P(|Y - \theta| \geq \epsilon) \to 0$ as $n \to \infty$.

For the normal population model, $\overline{X}$ is a consistent estimator of μ. This is because, given any $\epsilon > 0$,

$$P\big(|\overline{X} - \mu| \geq \epsilon\big) \leq \frac{\mathrm{Var}(\overline{X})}{\epsilon^2} = \frac{\sigma^2}{\epsilon^2} \cdot \frac{1}{n} \to 0.$$

This argument shows that for any model, with μ denoting the population mean, $\overline{X}$ is a consistent estimator of μ; it depends only on the fact that the variance of $\overline{X}$ is σ^2/n, where σ^2 is the (finite) population variance. In fact the same argument gives a general fact: if Y_n is an unbiased estimator (based on a sample of size n) of θ and if $\mathrm{Var}(Y_n) \to 0$, then Y_n is a consistent estimator of θ.

If Y is an unbiased estimator of θ, then of course $\mathbb{E}(Y - \theta)^2$ is nothing but the variance of Y. However, if Y is biased, then this is no longer the variance of Y; the quantity $\mathbb{E}(Y - \theta)^2$ is called the mean squared error (MSE). An estimator is said to have minimum MSE if this quantity is the least possible. When the estimator is unbiased, the MSE is its variance, and an unbiased estimator with minimum MSE is called the minimum variance unbiased estimator (MVUE). The sample mean $\overline{X}$ has minimum variance among all unbiased estimators that are linear combinations of $X_1, \ldots, X_n$.

2.6 Method of Moments: General Recipe and Normal Example

Method of moments. Let X be a random variable with density function $f(x; \theta_1, \theta_2)$, with parameters θ_1, θ_2 to be estimated, and let $X_1, \ldots, X_n$ be a random sample.

1. Calculate the first two sample moments $m_1 = \dfrac{1}{n}\sum_{i=1}^n X_i$ and $m_2 = \dfrac{1}{n}\sum_{i=1}^n X_i^2$.
2. Calculate the first two population moments $\mathbb{E}(X^k) = \int_{-\infty}^{\infty} x^k f(x; \theta_1, \theta_2)\, dx$; the results are in terms of θ_1 and θ_2.
3. Solve the simultaneous equations $\mathbb{E}(X) = m_1$, $\mathbb{E}(X^2) = m_2$ for θ_1, θ_2. The solutions are the method-of-moments estimators.

Example 2.15 (Normal distribution by the method of moments, revisited)**.** Consider $X \sim N(\mu, \sigma^2)$ with density

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

The parameters are μ and σ^2, and we have a random sample $X_1, \ldots, X_n$.

Step 1 (sample moments).

$$m_1 = \frac{1}{n}\sum_{i=1}^n X_i = \overline{X}, \qquad m_2 = \frac{1}{n}\sum_{i=1}^n X_i^2.$$

Step 2 (theoretical moments).

$$\hat{\mu}_1 = \mathbb{E}(X) = \mu, \qquad \hat{\mu}_2 = \mathbb{E}(X^2) = \mu^2 + \sigma^2, \qquad (\mathrm{Var}(X) = \sigma^2).$$

Step 3 (solve). From $\mu = \overline{X}$ and $\mu^2 + \sigma^2 = \frac{1}{n}\sum X_i^2$ we obtain

$$\hat{\mu} = \overline{X}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n X_i^2 - \overline{X}^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X})^2.$$

2.7 Maximum Likelihood with Numerical Data

Example 2.16 (ML estimate of μ from a binomial-type sample)**.** The following samples $0, 1, 0, 0, 1$ are from a binomial distribution which has the form $P(X = 0) = (1 - \mu)$ and $P(X = 1) = \mu$. Find the maximum likelihood estimate of μ.

The likelihood function is

$$L(\mu) = P(X{=}0)\, P(X{=}1)\, P(X{=}0)\, P(X{=}0)\, P(X{=}1) = (1-\mu)\,\mu\,(1-\mu)\,(1-\mu)\,\mu = (1-\mu)^4\mu^2.$$

Taking logarithms,

$$\log L(\mu) = \log\big[(1-\mu)^4\mu^2\big] = 4\log(1-\mu) + 2\log\mu.$$

Differentiating and equating to zero,

$$\begin{aligned}\frac{\partial \log L(\mu)}{\partial \mu} &= 4\cdot\frac{1}{1-\mu}(-1) + 2\cdot\frac{1}{\mu} = 0 \\ &\implies -4\mu + 2(1-\mu) = 0 \implies -6\mu + 2 = 0 \implies \mu = \frac{2}{6} = \frac{1}{3}.\end{aligned}$$

The second derivative is

$$\frac{\partial^2 \log L(\mu)}{\partial \mu^2} = \frac{-4}{(1-\mu)^2} + \frac{2}{\mu^2} < 0,$$

which is negative, confirming a maximum. Hence the maximum likelihood estimate is

$$\hat{\mu} = \frac{1}{3}.$$

Example 2.17 (ML estimate of a Poisson mean from numerical data)**.** The samples $5, 9, 3, 12, 14$ are from a Poisson distribution of the form $P(X = k) = \dfrac{\lambda^k e^{-\lambda}}{k!}$. What is the maximum likelihood estimator of λ? The likelihood function is

$$\begin{aligned}L(\lambda) &= P(X{=}5)P(X{=}9)P(X{=}3)P(X{=}12)P(X{=}14) \\ &= \frac{e^{-\lambda}\lambda^5}{5!}\cdot\frac{e^{-\lambda}\lambda^9}{9!}\cdot\frac{e^{-\lambda}\lambda^3}{3!}\cdot\frac{e^{-\lambda}\lambda^{12}}{12!}\cdot\frac{e^{-\lambda}\lambda^{14}}{14!} = \frac{e^{-5\lambda}\lambda^{43}}{5!\,9!\,3!\,12!\,14!}.\end{aligned}$$

Taking logarithms,

$$\log L(\lambda) = 43\log\lambda - 5\lambda - \log(5!\,9!\,3!\,12!\,14!).$$

Differentiating and equating to zero,

$$\frac{d\log L(\lambda)}{d\lambda} = 43\cdot\frac{1}{\lambda} - 5 = 0 \implies \lambda = \frac{43}{5}.$$

The second derivative $\dfrac{\partial^2 \log L(\lambda)}{\partial \lambda^2} = \dfrac{-43}{\lambda^2}$ is negative for any λ, confirming a maximum. Hence

$$\hat{\lambda} = \frac{43}{5}.$$

Chapter 3

Statistical Intervals for a Single Sample

3.1 Confidence, Tolerance, and Prediction Intervals

An interval estimate for a population parameter is called a *confidence interval*. We cannot be certain that the interval contains the true, unknown population parameter—we only use a sample from the full population to compute the point estimate and the interval. However, the confidence interval is constructed so that we have confidence that it does contain the unknown population parameter. Confidence intervals are widely used in engineering and the sciences.

Interval estimates can be of three types:

1. A **confidence interval** bounds population or distribution parameters (such as a mean viscosity).
2. A **tolerance interval** bounds a selected proportion of a distribution.
3. A **prediction interval** bounds future observations from the population or distribution.

3.2 Confidence Interval on the Mean, Variance Known

Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from a normal distribution with unknown mean μ and known variance σ^2. We know that the sample mean $\overline{X}$ is normally distributed with mean μ and variance σ^2/n. We may standardise $\overline{X}$ by subtracting the mean and dividing by the standard deviation, which results in the variable

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}.$$

The random variable Z has a standard normal distribution.

A confidence interval estimate for μ is an interval of the form $\ell \leq \mu \leq u$, where the endpoints ℓ and u are computed from sample data. Because different samples will produce different values of ℓ and u, these endpoints are values of random variables L and U respectively. Suppose that we can determine values of L and U such that the following probability statement is true:

$$P\{L \leq \mu \leq U\} = 1 - \alpha, \qquad 0 \leq \alpha \leq 1.$$

There is a probability of $1 - \alpha$ of selecting a sample for which the confidence interval will contain the true value of μ. The endpoints ℓ and u are called the *lower* and *upper confidence limits* respectively, and $1 - \alpha$ is called the *confidence coefficient*.

Because $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$ has a standard normal distribution, we may write

$$P\left\{-z_{\alpha/2} \leq \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right\} = 1 - \alpha.$$

Manipulating the terms inside the brackets by (1) multiplying through by $\sigma/\sqrt{n}$, (2) subtracting $\overline{X}$ from each term, and (3) multiplying through by -1, we obtain

$$P\left\{\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha.$$

Confidence Interval on the Mean, Variance Known

If $\overline{X}$ is the sample mean of a random sample of size n, with known variance σ^2, from a normal population, a $100(1-\alpha)\%$ confidence interval on μ is given by

$$\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}},$$

where $z_{\alpha/2}$ is the upper $100\,\alpha/2$ percentage point of the standard normal distribution.

Remark 3.1. For a fixed sample size n and standard deviation σ, the higher the confidence level, the longer the resulting confidence interval. The length of the confidence interval is a measure of the precision of estimation.

3.2.1 Choice of Sample Size

If $\overline{X}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\overline{X} - \mu|$ will not exceed a specified amount E when the sample size is

Sample Size for a Specified Error

$$n = \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2.$$

If the right-hand side of this equation is not an integer, it must be rounded up. This will ensure that the level of confidence does not fall below $100(1-\alpha)\%$. Notice that $2E$ is the length of the resulting confidence interval.

3.2.2 One-Sided Confidence Bounds

It is possible to obtain one-sided confidence bounds for μ by setting either $\ell = -\infty$ or $u = \infty$ and replacing $z_{\alpha/2}$ by z_α.

One-Sided Confidence Bounds on the Mean

A $100(1-\alpha)\%$ *upper* confidence bound for μ is

$$\mu \leq u = \overline{X} + z_\alpha\frac{\sigma}{\sqrt{n}},$$

and a $100(1-\alpha)\%$ *lower* confidence bound for μ is

$$\overline{X} - z_\alpha\frac{\sigma}{\sqrt{n}} = \ell \leq \mu.$$

3.3 Worked Problems: Mean, Variance Known

Example 3.1 (90% CI and sample size)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, **(i)** construct a 90% confidence interval on the population mean if the population standard deviation is known to be 2; **(ii)** determine the sample size required to keep the width of the confidence interval to less than 2.

(i) The $100(1-\alpha)\%$ confidence interval is $\mu \in \overline{X} \pm z_{0.05} \cdot \dfrac{\sigma}{\sqrt{n}}$. The sample mean is

$$\overline{X} = \frac{5+2+3+1+0+0+1+3+3}{9} = \frac{18}{9} = 2.$$

With $z_{0.05} = 1.645$, $\sigma = 2$, $n = 9$,

$$\text{CI} = 2 \pm 1.645 \cdot \frac{2}{\sqrt{9}} = 2 \pm 1.645 \cdot \frac{2}{3},$$

giving the upper limit $2 + 1.645 \cdot \frac{2}{3} = 3.0967$ and the lower limit $2 - 1.645 \cdot \frac{2}{3} = 0.9033$:

$$0.9033 \leq \mu \leq 3.0967.$$

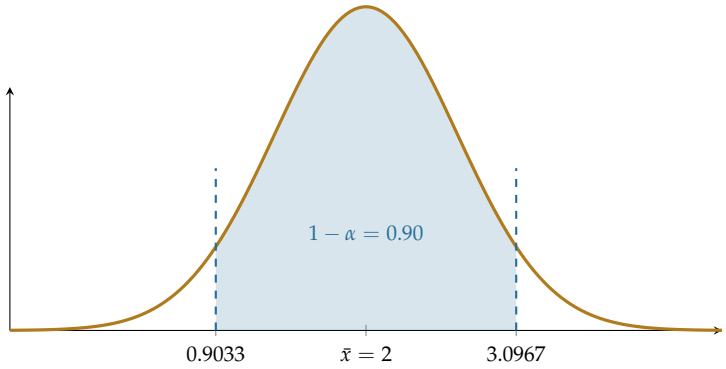


Figure 3.1: The 90% confidence interval $0.9033 \leq \mu \leq 3.0967$ centred at $\bar{x} = 2$, shown on the sampling distribution of the mean.

(ii) The width is

$$\text{width} = 2z_{0.05}\frac{\sigma}{\sqrt{n}} = \frac{2 \cdot 1.645 \cdot 2}{\sqrt{n}} < 2 \implies 2 \cdot 1.645 \cdot 2 < 2\sqrt{n} \implies \sqrt{n} > \frac{2 \cdot 1.645 \cdot 2}{2} = 3.29,$$

so $n \geq 10.8241$, i.e. $n \geq 11$.

Example 3.2 (95% upper bound; CI on variance)**.** For the sample $(5, 2, 3, 1, 4, 2, 1, 3, 3, 5)$ from a population representing the extent of overfill (mL) in sealed beverage bottles, **(a)** construct a 95% confidence *upper* bound on the population mean if the population variance is known to be 5 (what should the sample size be to keep the width of the confidence interval less than 1 mL?); **(b)** if the population variance is unknown, construct a 95% confidence interval on the population variance.

(a) 95% **upper bound on** μ**.** The sample mean is

$$\overline{X} = \frac{5+2+3+1+4+2+1+3+3+5}{10} = \frac{29}{10} = 2.9.$$

With $z_{0.05} \approx 1.645$,

$$\mu \leq \overline{X} + z_{0.05}\frac{\sigma}{\sqrt{n}} = 2.9 + 1.645 \cdot \frac{\sqrt{5}}{\sqrt{10}} = 2.9 + 1.645 \cdot \sqrt{\tfrac{5}{10}} = 4.063.$$

The upper bound is the same as the upper limit of the 90% two-sided confidence interval $\overline{X} \pm z_{0.05}\frac{\sigma}{\sqrt{n}} = 2.9 \pm 1.645\sqrt{\frac{5}{10}}$. For the width,

$$\text{width} = 2 \cdot 1.645\sqrt{\frac{5}{n}} < 1 \implies 2 \cdot 1.645\sqrt{5} < \sqrt{n} \implies n > (2 \cdot 1.645)^2 \cdot 5 = 54.1,$$

so the sample size $n \geq 54$ keeps the CI width < 1 mL.

(b) 95% **CI on σ^2.** With $\alpha = 0.05$, $n = 10$,

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} \implies \frac{9 \cdot 1.449^2}{19.02} \leq \sigma^2 \leq \frac{9 \cdot 1.449^2}{2.7},$$

where the sample variance is

$$s^2 = \frac{1}{9}\big[(5-2.9)^2 + (2-2.9)^2 + (3-2.9)^2 + (1-2.9)^2 + (4-2.9)^2 + (2-2.9)^2 + (1-2.9)^2 + (3-2.9)^2 + (3-2.9)^2 + (5-2.9)^2\big].$$

This gives

$$0.9937 \leq \sigma^2 \leq 7 \text{ mL}^2.$$

Example 3.3 (Standard error and a 99.7% interval)**.** From a population with known standard deviation of 1.65, a sample of 32 items resulted in 34.8 as an estimate of the mean. **(i)** Find the standard error of the mean. **(ii)** Compute an interval estimate that should include the population mean 99.7% of the time.

(i) The standard error of the mean is

$$\frac{\sigma}{\sqrt{n}} = \frac{1.65}{\sqrt{32}} = 0.29.$$

(ii) For a 99.7% interval, $z_{0.0015} = 2.97$:

$$\overline{X} \pm z_{0.0015}\frac{\sigma}{\sqrt{n}} = 34.8 \pm 2.97 \cdot \frac{1.65}{\sqrt{32}},$$

which gives

$$33.93 \leq \mu \leq 35.67.$$

Example 3.4 (Average burning time of light bulbs)**.** The manager of Cardinal Electricals' light-bulb division must estimate the average number of hours that a light bulb made by each bulb machine will last. A sample of 40 light bulbs was selected from a machine, and the average burning time was 1416 hours; the standard deviation of burning time is known to be 30 hours. Construct a 90% confidence interval for the true population mean. Here $n = 40$, $\overline{X} = 1416$, $\sigma = 30$, $\alpha = 0.1$:

$$\overline{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1416 \pm 1.645 \cdot \frac{30}{\sqrt{40}} = (1408.2,\ 1423.8).$$

$$1408.2 \leq \mu \leq 1423.8.$$

Example 3.5 (Operating-system response time: choosing n)**.** An operating system for a PC has been studied extensively, and it is known that the standard deviation of the response time following a particular command is $\sigma = 8$ milliseconds. A new version is installed, and we wish to estimate the mean response time for the new system to ensure that a 95% confidence interval for μ has length at most 5 milliseconds. **(a)** Assuming the response time is normally distributed with $\sigma = 8$, what sample size would you recommend? **(b)** Suppose the vendor tells us the standard deviation of the new system is smaller, $\sigma = 6$; give the recommended sample size and comment on the effect.

(a) With $\sigma = 8$, length $= 5$, confidence level 95% ($\alpha = 0.05$, $z_{0.025} = 1.96$),

$$\text{length} = 2z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 2 \cdot 1.96 \cdot \frac{8}{\sqrt{n}} = 5 \implies \sqrt{n} = \frac{2 \cdot 1.96 \cdot 8}{5} = 6.272,$$

so $n = (6.272)^2 = 39.338$. Therefore the sample size is

$$n = 40.$$

(b) With $\sigma = 6$,

$$\frac{2 \cdot 1.96 \cdot 6}{\sqrt{n}} = 5 \implies \sqrt{n} = 4.704 \implies n = (4.704)^2 = 22.13 \implies n = 23.$$

We can observe that as the standard deviation reduces, the required sample size also reduces.

Example 3.6 (Compressive strength of concrete: 95% and 99% CIs)**.** A civil engineer is analysing the compressive strength of concrete. Compressive strength is normally distributed with $\sigma^2 = 1000$ (psi)^2. A random sample of 12 specimens has a mean $\overline{X} = 3250$ psi. **(i)** Construct a 95% two-sided confidence interval on mean compressive strength. **(ii)** Construct a 99% two-sided confidence interval on mean compressive strength.

(i) 95% **CI** ($\alpha = 0.05$, $z_{0.025} = 1.96$):

$$\mu \in \overline{X} \pm z_{0.025}\frac{\sigma}{\sqrt{n}} = 3250 \pm 1.96 \cdot \frac{\sqrt{1000}}{\sqrt{12}},$$

$$3232.1 \leq \mu \leq 3267.89.$$

(ii) 99% **CI** ($\alpha = 0.01$, $z_{0.005} = 2.58$):

$$3250 - 2.58 \cdot \frac{\sqrt{1000}}{\sqrt{12}} \leq \mu \leq 3250 + 2.58 \cdot \frac{\sqrt{1000}}{\sqrt{12}},$$

$$3226.4 \leq \mu \leq 3273.55.$$

Example 3.7 (Recovering the sample mean and identifying the confidence level)**.** A random sample has been taken from a normal population and two confidence intervals constructed

using exactly the same data. The two confidence intervals are $(38.02, 61.98)$ and $(39.95, 60.05)$. **(a)** What is the value of the sample mean? **(b)** One of these intervals is a 90% CI and the other is a 95% CI. Which one is the 95% CI, and why?

(a) The confidence interval is $\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$, so $\overline{X}$ is the midpoint of the CI. Using either interval,

$$\overline{X} = \frac{38.02 + 61.98}{2} = \frac{100.00}{2} = 50.$$

$$\overline{X} = 50.$$

(b) For a fixed sample size n and standard deviation σ, the higher the confidence level, the longer the resulting confidence interval. We have

$$(39.95, 60.05) \Rightarrow \text{width} = 20.1 \;\rightarrow\; 90\%\text{ CI}, \qquad (38.02, 61.98) \Rightarrow \text{width} = 23.96 \;\rightarrow\; 95\%\text{ CI}.$$

The wider interval $(38.02, 61.98)$ is the 95% CI.

3.4 Confidence Interval on the Mean, Variance Unknown

When the sample is small (< 30 or < 40) and σ^2 is unknown, we must make an assumption about the form of the underlying distribution to obtain a valid CI procedure. A reasonable assumption is that the underlying distribution is normal.

Suppose the population of interest has a normal distribution with unknown mean μ and unknown variance σ^2. Assume that a random sample of size n, say $X_1, X_2, \ldots, X_n$, is available, and let $\overline{X}$ and S^2 be the sample mean and variance. To construct a two-sided CI on μ: if the variance were known, we would use $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$, which has a standard normal distribution. When σ^2 is unknown, a logical procedure is to replace σ by the sample standard deviation S. The random variable Z now becomes $T = (\overline{X} - \mu)/(S/\sqrt{n})$. (If n is large, proceed to use the CI based on the normal distribution itself.)

The t Distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with unknown mean μ and unknown variance σ^2. The random variable

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $n - 1$ degrees of freedom. Its probability density function is

$$f(x) = \frac{\Gamma[(k+1)/2]}{\sqrt{\pi k}\,\Gamma(k/2)} \cdot \frac{1}{\left[(x^2/k) + 1\right]^{(k+1)/2}}, \qquad -\infty < x < \infty,$$

where k is the number of degrees of freedom.

The mean and variance of the t distribution are zero and $k/(k-2)$ for $k > 2$, respectively. The general appearance of the t distribution is similar to the standard normal distribution: both are symmetric and unimodal, and the maximum ordinate value is reached when the mean $\mu = 0$. However, the t distribution has heavier tails than the normal; that is, it has more probability in the tails. As the number of degrees of freedom $k \to \infty$, the limiting form of the t distribution is

the standard normal distribution (Figure 3.2). Generally the number of degrees of freedom for t are the number of degrees of freedom associated with the estimated standard deviation.

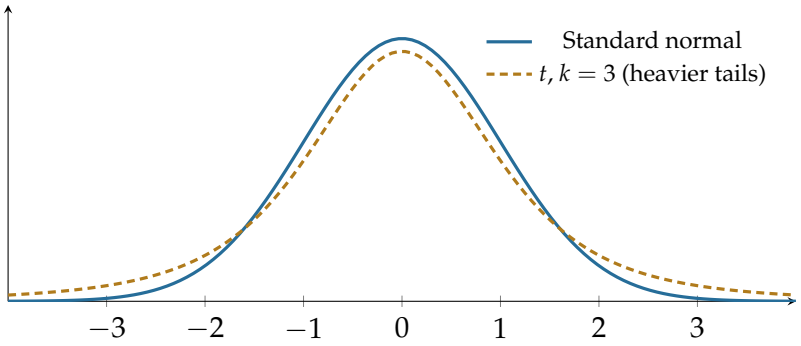


Figure 3.2: The t distribution (dashed) has heavier tails than the standard normal (solid). As $k \to \infty$ the t distribution approaches the standard normal.

The table of percentage points lets $t_{\alpha,k}$ be the value of the random variable T with k degrees of freedom above which we find an area (or probability) α. Thus $t_{\alpha,k}$ is an upper-tail $100\alpha\%$ point of the t distribution with k degrees of freedom. For example, the t value with 10 degrees of freedom having an area of 0.05 to the right is $t_{0.05,10} = 1.812$, i.e.

$$P(T_{10} > t_{0.05,10}) = P(T_{10} > 1.812) = 0.05.$$

Since the t distribution is symmetric about zero, we have $t_{1-\alpha,k} = -t_{\alpha,k}$; that is, the t value having an area of $1-\alpha$ to the right (an area of α to the left) is the negative of the t value that has area α in the right tail. Therefore $t_{0.95,10} = -t_{0.05,10} = -1.812$.

Confidence Interval on the Mean, Variance Unknown

If $\overline{X}$ and s are the mean and standard deviation of a random sample from a normal distribution with unknown variance σ^2, a $100(1-\alpha)\%$ confidence interval on μ is

$$\overline{X} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \le \mu \le \overline{X} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}},$$

where $t_{\alpha/2,n-1}$ is the upper $100\,\alpha/2$ percentage point of the t distribution with $n-1$ degrees of freedom.

One-sided confidence bounds on the mean of a normal distribution are obtained by using only the appropriate lower or upper confidence limit from the equation above and replacing $t_{\alpha/2,n-1}$ by $t_{\alpha,n-1}$. It is not as easy to select a sample size n to obtain a specified length for this CI as it was in the σ-known case, because the length of the interval involves s, n, and $t_{\alpha/2,n-1}$, and the t-percentile itself depends on n. Consequently an appropriate n can only be obtained through trial and error.

3.5 Worked Problems: Mean, Variance Unknown

Example 3.8 (Finding t-percentiles)**.** Determine the t-percentile required to construct each of the following.

(a) Two-sided CI, confidence level 95%, dof = 12: $t_{\alpha/2,n-1} = t_{0.025,12} = 2.179$.

(b) Two-sided CI, confidence level 99.9%, dof = 15: $t_{\alpha/2,n-1} = t_{0.0005,15} = 4.073$.

(c) One-sided CI, confidence level 99%, dof $= 19$: $t_{\alpha,n-1} = t_{0.01,19} = 2.539$.
(d) One-sided CI, confidence level 99.9%, dof $= 24$: $t_{\alpha,n-1} = t_{0.001,24} = 3.467$ (for dof $= 15$, $t_{0.001,15} = 3.733$).

Example 3.9 (Tire life)**.** A research engineer for a tire manufacturer is investigating tire life for a new rubber compound and has built 16 tires, testing them to end of life in a road test. The sample mean and standard deviation are 60139.7 and 3645.94 km. Find a 95% confidence interval on mean tire life μ. With $\alpha = 0.05$, $n = 16$, $t_{0.025,15} = 2.131$,

$$\overline{X} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \le \mu \le \overline{X} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 60139.7 \pm 2.131 \cdot \frac{3645.94}{\sqrt{16}},$$

$$58197.33 \le \mu \le 62082.07.$$

Example 3.10 (Electronic sensor life: known vs. unknown variance)**.** In a maintenance department of a firm, the life of the electronic equipment is found to be normally distributed. A random sample of 30 sensors has a sample mean life of 800 hours and a sample standard deviation of 70 hours. **(i)** Construct a 95% two-sided CI for the mean life of the sensors. **(ii)** Construct a 99% two-sided CI for the variance of the life of sensors. **(iii)** If the population standard deviation is known to be 90 hours, construct a 99% two-sided CI for the mean life of the sensors. **(iv)** Comment on the difference in answers to (i) and (iii).

Here Life $\sim N(\mu, \sigma^2)$, $\overline{X} = 800$, $s = 70$, $n = 30$.

(i) $\alpha = 0.05$, $t_{0.025,29} = 2.045$:

$$\overline{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 800 \pm 2.045 \cdot \frac{70}{\sqrt{30}} = 800 \pm 26.135,$$

$$773.86 \le \mu \le 826.14 \text{ hrs.}$$

(ii) CI on variance.

$$\frac{(n-1)s^2}{\chi^2_{0.005,29}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{0.995,29}} = \frac{29 \cdot 70^2}{52.34} \text{ and } \frac{29 \cdot 70^2}{13.12},$$

$$2715 \le \sigma^2 \le 10{,}830.7.$$

(iii) σ **known** $= 90$**.**

$$\overline{X} \pm z_{0.005}\frac{\sigma}{\sqrt{n}} = 800 \pm 2.58 \cdot \frac{90}{\sqrt{30}} = 800 \pm 42.39,$$

$$757.6 \le \mu \le 842.4.$$

(iv) The confidence interval of (iii) is wider than that of the 99% CI in spite of the population variance being known.

Example 3.11 (Electronic sensor life: a second data set)**.** The life of an electronic sensor is normally distributed. A random sample of 20 sensors has a mean life of 1000 hours and a standard deviation of 100 hours. **(i)** Construct a 99% two-sided CI for the mean life of the sensors. **(ii)** Construct a 95% two-sided CI for the variance of the life of sensors. **(iii)** If the population standard deviation is known to be 105 hours, construct a 99% two-sided CI for the mean life of the sensors. **(iv)** Comment on the difference in answers to (i) and (iii).

Here Life $\sim N(\mu, \sigma^2)$, $\overline{X} = 1000$, $s = 100$, $n = 20$.

(i) 99% CL, $\alpha = 0.01$, $t_{0.005,19} = 2.861$:

$$\overline{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 1000 \pm 2.861 \cdot \frac{100}{\sqrt{20}},$$

$$936.03 \leq \mu \leq 1063.97 \text{ hrs.}$$

(ii) $\alpha = 0.05$, $\alpha/2 = 0.025$, $\chi^2_{0.025,19} = 32.85$, $\chi^2_{0.975,19} = 8.91$:

$$\frac{(n-1)s^2}{\chi^2_{0.025,19}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{0.975,19}} = \frac{19 \cdot 100^2}{32.85} \text{ and } \frac{19 \cdot 100^2}{8.91},$$

$$5783.87 \leq \sigma^2 \leq 21{,}324.35.$$

(iii) $\sigma = 105$ known, $\alpha = 0.01$:

$$\overline{X} \pm z_{0.005}\frac{\sigma}{\sqrt{n}} = 1000 \pm 2.58 \cdot \frac{105}{\sqrt{20}},$$

$$939.42 \leq \mu \leq 1060.58 \text{ hrs.}$$

(iv) The CI of (iii) is narrower compared to (i) since σ is known. A more precise interval is obtained when additional information is available.

Example 3.12 (95% CI on the mean, variance unknown)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, construct a 95% confidence interval on the population mean if the population variance is unknown. The sample mean is $\overline{X} = 2$, and the sample standard deviation is computed from

$$\begin{aligned} s^2 &= \frac{1}{8}\Big[(5-2)^2 + (2-2)^2 + (3-2)^2 + (1-2)^2 + (0-2)^2 \\ &\quad + (0-2)^2 + (1-2)^2 + (3-2)^2 + (3-2)^2\Big] \\ &= \frac{9+0+1+1+4+4+1+1+1}{8} = \frac{22}{8} = 2.75, \end{aligned}$$

so $s = 1.658$. With $t_{0.025,8} = 2.306$,

$$\mu \in \overline{X} \pm t_{0.025,8}\frac{s}{\sqrt{n}} = 2 \pm 2.306 \cdot \frac{1.658}{\sqrt{9}},$$

$$0.7253 \le \mu \le 3.2747.$$

Example 3.13 (Cloud seeding: rainfall in acre-feet)**.** Cloud seeding has been studied for many decades as a weather modification procedure. The rainfall in acre-feet from 20 clouds that were selected at random and seeded with silver nitrate are as follows:

18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, 31.6.

(a) Construct a 95% one-sided lower bound on the mean. **(b)** Construct a 90% two-sided CI on the mean. What is the width of this interval? **(c)** If the width of the interval in part (b) were to be halved, how much should the sample size be? Use trial and error if needed.

Here $n = 20$, $s = 4.7848$, $\overline{X} = 26.035$.

(a) 95% **lower bound.** With $t_{0.05,19} = 1.729$,

$$\mu \ge \overline{X} - t_{0.05,19}\frac{s}{\sqrt{n}} = 26.035 - 1.729 \cdot \frac{4.7848}{\sqrt{20}},$$

$$\mu \ge 24.185.$$

(b) 90% **two-sided CI.** With $\alpha = 0.10$, $t_{0.05,19} = 1.729$,

$$\overline{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 26.035 \pm 1.729 \cdot \frac{4.7848}{\sqrt{20}},$$

$$90\%\ \text{CI} = (24.185,\ 27.895), \qquad \text{width} = 27.895 - 24.185 = 3.7958.$$

(c) Halving the width. The width is $2t_{0.05,\,n-1}\dfrac{s}{\sqrt{n}}$ (assuming s remains the same). By trial and error,

$$n = 30 \Rightarrow \text{width} = 2 \cdot 1.697 \cdot \tfrac{4.7848}{\sqrt{30}} = 2.96,$$
$$n = 40 \Rightarrow \text{width} = 2 \cdot 1.684 \cdot \tfrac{4.7848}{\sqrt{40}} = 2.5,$$
$$n = 60 \Rightarrow \text{width} = 2 \cdot 1.671 \cdot \tfrac{4.7848}{\sqrt{60}} = 2.06,$$
$$n = 120 \Rightarrow \text{width} = 2 \cdot 1.658 \cdot \tfrac{4.7848}{\sqrt{120}} = 1.4484.$$

Half the width of part (b) is $3.7958/2 = 1.8979$. Therefore n should be between 60 and 120.

Example 3.14 (Gasoline prices and highway speeds)**. (a)** From a sample of 42 gasoline stations statewide, the average price of a gallon of unleaded gas was found to be \$1.12 and the standard deviation was \$0.04 per gallon. Construct a 99% CI on the mean price of a gallon of unleaded gas. **(b)** A random sample of 57 vehicles on a highway clocked an average of 23.2 mph with a standard deviation of 0.3 mph. **(i)** Estimate the standard deviation of the population. **(ii)** Estimate the standard error of the mean for this population.

(a) $n = 42$, $\overline{X} = 1.12$, $s = 0.04$, $\alpha = 0.01$, $t_{0.005,41} = 2.7$:

$$\overline{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} = 1.12 \pm 2.7 \cdot \frac{0.04}{\sqrt{42}},$$

$$99\%\ \text{CI} = [\$1.10,\ \$1.14].$$

(b) $n = 57$, $\overline{X} = 23.2$, $s = 0.3$, assume $\alpha = 0.05$.
(i) Using the CI on the variance,

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} \right) = \left(\frac{56 \cdot 0.3^2}{77.4}, \frac{56 \cdot 0.3^2}{36.5} \right),$$

so the 95% CI for σ^2 is $(0.065, 0.138)$, and the CI for σ is $(0.255, 0.37)$.
(ii) The standard error of the mean is

$$\frac{\hat{\sigma}}{\sqrt{n}} = \frac{0.255 + 0.37}{2} \cdot \frac{1}{\sqrt{57}} = 0.041.$$

Example 3.15 (Contamination level in food packs)**.** The level of contamination (in %) was studied in 6 packs of food with the following data: 16.8, 17.2, 17.4, 16.9, 16.5, 17.1. Find a 99% confidence interval on the mean % μ and provide a practical interpretation. The sample mean is

$$\overline{X} = \frac{101.9}{6} = 16.98,$$

and the sample variance is

$$s^2 = \frac{0.0324 + 0.0484 + 0.1764 + 0.006 + 0.2304 + 0.0144}{5} = 0.1016, \qquad s = 0.318.$$

With $t_{0.005,5} = 4.032$,

$$\overline{X} \pm t_{\alpha/2,5} \frac{s}{\sqrt{n}} = 16.98 \pm 4.032 \cdot \frac{0.318}{\sqrt{6}} = 16.98 \pm 0.523,$$

$$16.457 \leq \mu \leq 17.503.$$

Example 3.16 (Wall thickness of glass bottles)**.** The wall thickness of 25 glass bottles was measured. The sample mean was 4.05 mm and the sample standard deviation was 0.08 mm. Find a 95% lower confidence bound for mean wall thickness, and interpret the interval. Here $n = 25$, $\overline{X} = 4.05$, $s = 0.08$, $\alpha = 0.05$, $t_{0.05,24} = 1.711$:

$$\mu \geq \overline{X} - t_{\alpha,n-1} \frac{s}{\sqrt{n}} = 4.05 - 1.711 \cdot \frac{0.08}{\sqrt{25}},$$

$$\mu \geq 4.02 \text{ mm}.$$

The minimum population mean wall thickness is 4.02 mm.

3.6 Confidence Interval on the Variance and Standard Deviation

The χ^2 Distribution

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2, and let S^2 be the sample variance. Then the random variable

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

has a chi-square (χ^2) distribution with $n-1$ degrees of freedom.

Confidence Interval on the Variance

If s^2 is the sample variance from a random sample of n observations from a normal distribution with unknown variance σ^2, then a $100(1-\alpha)\%$ confidence interval on σ^2 is

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}},$$

where $\chi^2_{\alpha/2,n-1}$ and $\chi^2_{1-\alpha/2,n-1}$ are the upper and lower $100\,\alpha/2$ percentage points of the chi-square distribution with $n-1$ degrees of freedom. A confidence interval for σ has lower and upper limits that are the square roots of the corresponding limits for σ^2.

One-Sided Confidence Bounds on the Variance

The $100(1-\alpha)\%$ lower and upper confidence bounds on σ^2 are

$$\frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} \leq \sigma^2 \qquad \text{and} \qquad \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}}.$$

3.7 Worked Problems: Variance and Standard Deviation

Example 3.17 (Reading χ^2 percentiles)**.** Determine the values of the following percentiles:

$$\text{(a) } \chi^2_{0.05,10} = 18.31, \qquad \text{(b) } \chi^2_{0.995,16} = 5.14, \qquad \text{(c) } \chi^2_{0.005,25} = 46.93.$$

Example 3.18 (Selecting χ^2 percentiles for CIs)**.** Determine the χ^2 percentile required to construct each of the following CIs.

(a) Confidence level 95%, dof $= 24$, one-sided upper: $\chi^2_{0.05,24} = 36.42$.
(b) Confidence level 99%, dof $= 9$, one-sided upper: $\chi^2_{1-0.01,9} = \chi^2_{0.99,9} = 2.09$.
(c) Confidence level 90%, dof $= 19$, two-sided: $\chi^2_{0.1/2,19} = \chi^2_{0.05,19} = 30.14$.

Example 3.19 (Rivet hole diameter: 99% lower bound on variance)**.** A rivet is to be inserted into a hole. A random sample of $n = 15$ parts is selected, and the hole diameter is measured. The sample standard deviation of the hole-diameter measurements is $s = 0.008$ mm. Construct a 99% lower confidence bound for the variance. With $n = 15$ and $\chi^2_{0.01,14} = 29.14$,

$$\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{0.01,n-1}} = \frac{14 \cdot 0.008^2}{29.14} \geq 3.074 \times 10^{-5},$$

$$\text{CI} = [\,0.00003074,\ \infty\,).$$

Example 3.20 (Sugar content of canned peaches)**.** The sugar content of the syrup in canned peaches is normally distributed. A random sample of $n = 10$ cans yields a sample standard deviation of $s = 4.8$ mg. Calculate a 95% two-sided CI for σ. With $\alpha = 0.05$, $\chi^2_{0.025,9} = 19.02$, $\chi^2_{0.975,9} = 2.70$,

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} = \frac{9 \cdot 4.8^2}{19.02} \le \sigma^2 \le \frac{9 \cdot 4.8^2}{2.70},$$

so $10.90 \le \sigma^2 \le 76.8$. Taking square roots, $\sqrt{10.9} \le \sigma \le \sqrt{76.8}$:

$$3.30 \le \sigma \le 8.76.$$

Example 3.21 (95% CI on variance for a small sample)**.** For the sample $(5, 2, 3, 1, 0, 0, 1, 3, 3)$ from a population, construct a 95% confidence interval on the population variance. Here $s = 1.6583$ and $n = 9$, so with $\chi^2_{0.025,8} = 15.51$ and $\chi^2_{0.975,8} = 2.73$,

$$\frac{(n-1)s^2}{\chi^2_{0.025,8}} \le \sigma^2 \le \frac{(n-1)s^2}{\chi^2_{0.975,8}} = \frac{8 \cdot 1.6583^2}{15.51} \le \sigma^2 \le \frac{8 \cdot 1.6583^2}{2.73},$$

giving $1.4189 \le \sigma^2 \le 8.058$, and taking square roots,

$$1.19 \le \sigma \le 2.838.$$

Example 3.22 (Amino-acid concentration: variance test and one-sided bounds)**.** Amino-acid concentrations in 8 packs of dog feed were recorded in mg/kg as follows: $22, 24, 20, 26, 27, 24, 26, 23$. **(i)** Perform a test at 5% significance to see if the population variance is greater than 5.5 $(\text{mg/kg})^2$. What is the p-value of the test? **(ii)** Construct upper and lower 90% confidence bounds for the population variance.

(i) $H_0 : \sigma^2 = 5.5$, $H_1 : \sigma^2 > 5.5$ $(\text{mg/kg})^2$, $\alpha = 0.05$, $n = 8$, $\overline{X} = 24$, $s = 2.3299$. The sample variance is

$$s^2 = \frac{4 + 0 + 16 + 4 + 9 + 0 + 4 + 1}{7} = \frac{38}{7} = 5.428, \qquad s = \sqrt{5.428} = 2.3299.$$

The test statistic is

$$\chi^2_0 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{7 \cdot 2.3299^2}{5.5} = 6.9091.$$

The tabulated value is $\chi^2_{1-0.05,7} = \chi^2_{0.05,7} = 14.07$. Since $\chi^2_0 = 6.9091 < 14.07$, we do not reject H_0 (Figure 3.3). The p-value is $P(\chi^2_7 > 6.91)$, which lies between 0.4 and 0.5, since $P(\chi^2_7 > 6.346) = 0.5$ and $P(\chi^2_7 > 7.283) = 0.4$.
(ii) 90% **bounds.** The 90% *upper* bound is

$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}} = \frac{7 \cdot 5.4286}{2.833} = 13.41,$$

and the 90% *lower* bound is

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} = \frac{7 \cdot 5.4286}{12.02} = 3.16,$$

that is,

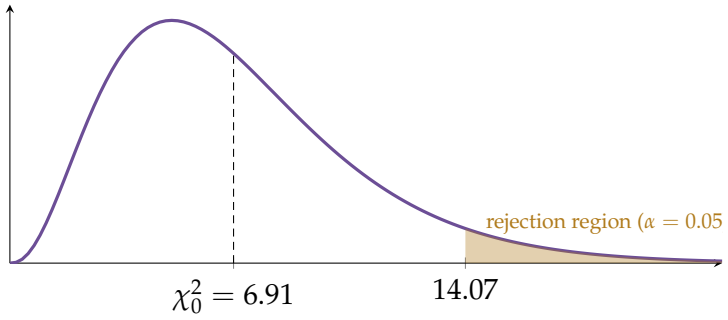


Figure 3.3: Chi-square density with 7 degrees of freedom. The observed statistic $\chi^2_0 = 6.91$ falls well short of the critical value 14.07, so H_0 is not rejected.

$$3.16 \le \sigma^2 \le 13.41.$$

Example 3.23 (Sugar content (continued): standard-deviation bounds)**.** For the canned-peaches data ($n = 10$, $s = 4.8$): **(a)** Find a 90% two-sided confidence interval for the population standard deviation. **(b)** Find a 95% upper bound and a 95% lower bound for the population standard deviation. How do these bounds compare with the limits of part (a)?

(a) $\alpha = 0.1$, two-sided $\alpha/2 = 0.05$, $\chi^2_{0.05,9} = 16.92$, $\chi^2_{0.95,9} = 3.325$:

$$\left[\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}\right] = \left[\frac{9 \cdot 4.8^2}{16.92}, \frac{9 \cdot 4.8^2}{3.325}\right] = [12.26,\ 62.36].$$

The interval for σ is $(3.5, 7.9)$.

(b) 95% upper bound ($\alpha = 0.05$, $\chi^2_{0.95,9} = 3.325$):

$$\sigma^2 \le \frac{(n-1)s^2}{\chi^2_{1-\alpha,n-1}} = \frac{9 \cdot 4.8^2}{3.325} = 62.36 \implies \sigma \le 7.9.$$

95% lower bound ($\chi^2_{0.05,9} = 16.92$):

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} = \frac{9 \cdot 4.8^2}{16.92} = 12.26 \implies \sigma \ge 3.5.$$

Both bounds equal the limits of the two-sided CI, since $\alpha/2$ in part (a) is the same as α in part (b).

Example 3.24 (Hole diameter: 99% lower bound on variance)**.** A hole diameter is measured in a random sample of 15 parts. The sample standard deviation of the hole diameter is 0.008 mm. Construct a 99% lower confidence bound for σ^2. With $n = 15$, $\alpha = 0.01$, $\chi^2_{0.01,14} = 29.14$,

$$\sigma^2 \ge \frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} = \frac{14 \cdot 0.008^2}{29.14} = 3.84 \times 10^{-3},$$

$$\sigma \ge 0.062.$$

Example 3.25 (Weight measurements: test on σ and a lower bound)**.** A set of 15 weight measurements gives a sample standard deviation $s = 0.0089$ g. **(i)** Does the population standard deviation σ of all weight measurements exceed 0.008 g at 5% significance? Find the p-value for this test. **(ii)** Answer part (i) with the help of a lower confidence interval for σ with $\alpha = 5\%$.

Here $n = 15$, $s = 0.0089$ g.

(i) $H_0 : \sigma = 0.008$ g, $H_1 : \sigma > 0.008$ g, $\alpha = 0.05$. The test statistic is

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{14 \cdot 0.0088^2}{0.008^2} = 16.94.$$

The tabulated value is $\chi^2_{0.05,14} = 23.68$. Since $\chi_0^2 = 16.94 < \chi^2_{\text{table}} = 23.68$, we fail to reject the null hypothesis; the standard deviation does not exceed 0.008 g at 5% significance. For the p-value, 16.94 lies between the table entries for 0.5 and 0.1 (21.06 at $p = 0.1$ and 13.34 at $p = 0.5$), so $0.1 < p < 0.5$.

(ii) Lower confidence bound, $\alpha = 0.05$, $\chi^2_{0.05,14} = 23.68$:

$$\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{0.05,14}} = \frac{14 \cdot 0.0088^2}{23.68} = 4.578 \times 10^{-5} \implies \sigma \geq 6.76 \times 10^{-3} \approx 0.0067.$$

$$\sigma \geq 0.0067.$$

Example 3.26 (Titanium percentage in an aerospace alloy)**.** The percentage of titanium in an alloy used in aerospace casting is measured in 51 randomly selected parts. The sample standard deviation is $s = 0.28$. Construct a 95% two-sided confidence interval for σ. With $\alpha = 0.05$, $n = 51$, $\chi^2_{0.025,50} = 71.42$, $\chi^2_{0.975,50} = 32.36$,

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}} = \frac{50 \cdot 0.28^2}{71.42} \leq \sigma^2 \leq \frac{50 \cdot 0.28^2}{32.36},$$

so $0.055 \leq \sigma^2 \leq 0.121$. Taking the square root of the endpoints,

$$0.234 \leq \sigma \leq 0.348.$$